KinetiQ

UI Integration Points

Milestone 2 · Lifecycle Architecture · PEI 2026/2027, Universidade de Aveiro

The KinetiQ user interfaces (the React web back-office and the Flutter mobile app) reach the backend through a single entry point: the **Reverse Proxy (Nginx)** followed by the **API Gateway**. Requests travel over **HTTPS** (REST) and live data is pushed over **WSS** (WebSocket). The UIs never access the MQTT Broker, Kafka or the databases directly (NFR10, NFR23).

Integration points

#	Integration point	Channel	Path through the backend	Data exchanged
1	Authentication	HTTPS	UI → Reverse Proxy → API Gateway → Authentication Service (Keycloak). The Gateway checks the access token with Keycloak on every request.	Login and access token
2	Fleet and bike management	HTTPS	UI → Reverse Proxy → API Gateway → Core API → Fleet Management Service → PostgreSQL	Bike list and details, status, registration, lifecycle, maintenance tickets
3	Zones	HTTPS	UI → Reverse Proxy → API Gateway → Core API → Fleet Management Service → PostgreSQL/PostGIS. Zone changes are published to Kafka ("zone updates").	View and manage geographic zones
4	Live telemetry and events	WSS	Kafka → WebSocket Service → API Gateway → Reverse Proxy → UI	Real-time bike data, status changes and alerts, pushed to the map and dashboards
5	Telemetry history	HTTPS	UI → Reverse Proxy → API Gateway → Core API → InfluxDB	Time-series data by bike and time range
6	Value-added modules	HTTPS	UI → Reverse Proxy → API Gateway → Core API ↔ Value-added Modules (ML and BI)	Analytics results, reports, forecasts and anomaly flags

Integration points by actor

Permissions are enforced per role (OPERATOR, TECHNICIAN, ANALYST, RIDER) through RBAC in Keycloak (FR24), and each interface only shows the features its role is allowed to use (NFR08). The Maintenance Technician and the Fleet Analyst inherit the Fleet Operator's view of the fleet, limited by their role.

Actor	Points used	What for
Fleet Operator	1, 2, 3, 4, 5, 6	Full fleet management: bike lifecycle, users and modules, zones, live map, history and module results
Maintenance Technician	1, 2, 4, 5	Receives alerts live (4), claims tickets and updates the maintenance state (2), checks the bike's history (5)
Fleet Analyst	1, 2, 5, 6	Read-only fleet view (2), historical telemetry (5), BI reports, demand forecasts, anomaly flags and fleet CO_2 (6)
Rider (mobile app)	1, 2, 4, 5	Associates with a bike by QR code and reports faults (2), follows the current trip live (4), consults trip history (5)

Notes

- The web app and the mobile app use the same integration points; only the features shown differ by role.
- Live data (point 4) is delivered by the WebSocket Service and does not go through the Core API. Stored data and business operations (points 2, 3, 5 and 6) go through the Core API.
- All external traffic is encrypted: HTTPS (TLS 1.3) for requests and WSS for live data (NFR19).
- Still to be confirmed: the integration point for CO_2 savings (fleet and personal) depends on where the CO_2 calculation runs, which has not been decided yet.